

SAUMALYA SARKAR

Results-driven professional targeting roles in **Data Engineering | Data Warehousing | Lakehouse Development | LLMOps** to leverage expertise in AWS, Databricks, Delta Lake, and scalable data architectures to drive business growth through optimized, high-performance data solutions, preferably in Kolkata and Hyderabad.

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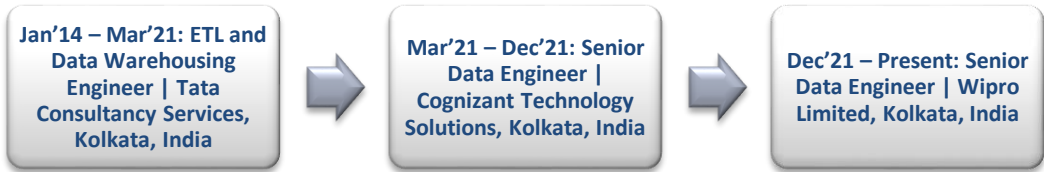
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PROFILE SUMMARY

- AWS and Databricks Certified Data Engineer** with over **11 years** of expertise in **Cloud Technologies and Data Warehousing**, focused on driving innovative business solutions.
- Specialized in **Delta Lake and AWS technologies**, leveraging tools like **AWS Glue, S3, Redshift, and Lambda** to build scalable, efficient data processing systems.
- Strong background in **designing and implementing ETL pipelines, data migration, and cloud architecture optimization** for performance and cost efficiency.
- Expertise in big data technologies like Apache Spark, PySpark and Informatica, delivering high-quality, performance-driven data processing pipelines.
- Proven track record of **migrating legacy data systems** to near-real-time solutions, reducing data latency from days to minutes.
- Expertise in **implementing processes and systems** to monitor **data quality**, ensuring production data is always accurate and available for key stakeholders and business processes that depend on it.
- Displayed success in **reducing operational costs (up to 30%)** and enhancing **data processing speed (up to 80%)** through cloud-native solutions and automation.
- Skilled in **Infrastructure-as-Code (IaC)**, implementing solutions using **Terraform, GitHub Actions, and CloudFormation** to standardize deployments and minimize errors.
- Proficient in **data warehousing and BI reporting**, with a focus on reducing report generation time by up to **60%** and improving decision-making processes.
- Hands-on experience in building AI stack using LlamaIndex, LangGRAPH, Ollama, vLLM, Huggingface.

CAREER TIMELINE



WORK EXPERIENCE

Dec'21 – Present: Senior Data Engineer | Wipro Limited, Kolkata, India

Key Result Areas:

- Completed multiple End-To-End AI application POC using LlamaIndex, LangGraph, Langfuse, vLLM and Huggingface. Some of those applications are in development phase now.
- Migrating legacy ETL processes to Near-Real-Time data marts, reducing data latency from days to mere minutes.
- Utilizing PySpark for advanced data processing and analytics to drive business insights.

Highlights:

- Designed and implemented a scalable Delta Lake solution leveraging Apache Hudi, AWS Glue Catalog, and S3 that resulted in a **50%-80%** improvement in data processing speed.
- Developed ETL pipelines using AWS Glue, S3, Lambda, and Redshift and achieved **99.9%** data accuracy.
- Contributed to the development of Infrastructure-as-Code (IaC) workflow with Terraform and GitHub Actions, while ensuring standardized deployments across **3+** environments and eliminating manual configuration errors.
- Managed and processed **150 TB** of data weekly, thereby leveraging cloud-native technologies for optimization and scalability.
- Architected a scalable BI infrastructure with AWS Data Pipelines, Redshift, and ThoughtSpot, which resulted in a **60%** reduction in report generation time, enhancing data-driven decision-making.

Mar'21 – Dec'21: Senior Data Engineer | Cognizant Technology Solutions, Kolkata, India

Key Result Areas:

CORE COMPETENCIES

Cloud Architecture Design

Cloud Migration & Optimization

Lakehouse Design & Development

Data Warehousing

Distributed Data Processing

Performance Optimization

Business Intelligence

LLMOps, Llamaindex, LangGraph

Ollama,vLLM,Huggingface,OpenAI,Cuda

Sagemaker, Bedrock

TECHNICAL SKILLS

- Programming Language:** Python 2.7-3.9, Shell Scripting
- Data Processing & Analytics** AWS Glue, Apache, Spark, Informatica, SQL, Thoughtspot, Azure Data Factory
- Databases** AWS Redshift, IBM Netezza, Google Big Query, NoSQL, Postgres, Aurora, RDBMS AMPP
- AWS** Glue S3, Lambda, EC2, IAM Identity Center, Step Function, Glue Catalog, DynamoDB, Sagemaker, Bedrock
- LLMOps** OpenAI, Ollama, LlamIndex, vLLM, LangGraph
- Databricks** Delta Table, Delta.io, Delta Lake, Unity Catalog
- Additional Skills** GitHub Actions/Workflow, Terraform, Docker, SSO, Microsoft Entra ID

CERTIFICATIONS

- AWS Certified Data Analytics - Specialty
- Databricks Certified Data Engineer - Associate
- Databricks Certified Data Engineer - Professional
- IBM PureData Systems for Analytics v7.1

EDUCATION

- Built an internal deployment tool using S3, EventBridge, and CloudFormation, automating processes and improved efficiency.
- Orchestrated & scheduled data pipelines with Apache Airflow thus enabling seamless workflow management.
- Contributed to Airflow microservices hosting framework with Docker, AWS ECS, and EC2, thereby improving application scalability and efficiency.

Highlights:

- Developed and optimized event-driven data processing pipelines, that led to a **15%** reduction in AWS operational costs.
- Designed and implemented a serverless data mart on AWS, thus enhancing performance and scalability while reducing storage costs by **20%-30%**.
- Improved data processing efficiency by **25%** through the use of AWS Glue for job management and PySpark implementations.
- Optimized data storage by implementing efficient compression algorithms, that resulted in a **20%** reduction in data storage costs.

Jan’14 – Mar’21: ETL and Data Warehousing Engineer | Tata Consultancy Services, Kolkata, India

Key Result Areas:

- Spearheaded the development of a Data Warehouse using Informatica PowerCenter and IBM Netezza for a high-revenue client, thus enabling advanced data analysis.
- Led a cloud migration proof-of-concept (POC) for the client, designed and implemented strategies to streamline the transition.
- Enhanced cost analysis effectiveness during the cloud migration process and drove informed decision-making and better resource allocation.

Highlights:

- Reduced management overhead by **30%** by replacing traditional server infrastructure with AWS services.
- Increased ROI by **15%** by establishing effective monitoring practices during the cloud migration.
- Modeled and designed over **10** data marts tailored to specific BI reporting needs, while ensuring optimized data structures.
- Optimized the performance of **100+** data pipelines, thereby improving ETL architecture and reducing processing time by **20%**.

- **2013: B.Tech. in Electronics & Communication** from West Bengal University of Technology, Kolkata, India



SOFT SKILLS

- Effective Communication
- Analytical Thinking
- Technical Adaptability
- Multitasking
- Team Management

PERSONAL DETAILS

Date of Birth:	24 th October 1991
Languages Known:	English, Hindi, Bengali
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